The Invisible Singularity:

Artificial Intelligence as Convergent Infrastructure and the Reorganization of Global Work, 2027–2035

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Abstract

The contemporary artificial intelligence landscape is characterized by an extraordinary cadence of model releases — successive frontier systems appearing at intervals of weeks rather than years. This paper argues that this proliferation is not the terminal state of the technology but a transitional phase, and that the observable trajectory points toward a structural convergence: a small number of highly capable, general-purpose AI systems that recede into the infrastructural background of economic life, invoked as unremarkably as search engines or enterprise resource platforms are invoked today. Drawing on the economics of general purpose technologies (Bresnahan & Trajtenberg, 1995), the theory of ubiquitous computing (Weiser, 1991), and the historical dynamics of technological revolutions (Perez, 2002; David, 1990), I develop a three-phase model — Proliferation, Acceleration, and Invisibility — and advance two central hypotheses. First, the Acceleration Hypothesis: beginning approximately in 2027, the rate of capability diffusion will compress such that materially novel tools and applications emerge on a near-daily cadence, rendering pre-2024 workflows structurally obsolete across most knowledge-intensive sectors. Second, the Reinstatement Hypothesis: contrary to popular displacement narratives, the task-creation channel identified by Acemoglu and Restrepo (2019) will dominate, producing net employment growth concentrated in newly instantiated occupational categories, consistent with the World Economic Forum's (2025) projection of 170 million jobs created against 92 million displaced by 2030. The paper's contribution is a unified framework that explains why the most transformative outcome of the current AI race is not a visible superintelligence but an invisible one — a utility so deeply embedded that, like electricity, its presence is noticed only in its absence. Limitations, falsifiable predictions, and counterarguments are addressed.

Keywords: *artificial intelligence, general purpose technology, technological singularity, future of work, infrastructural convergence, labor economics, foundation models*

1. Introduction

Between late 2022 and mid-2026, the field of artificial intelligence transitioned from a research discipline into a consumer-facing industrial race. Frontier laboratories now ship successive generations of large-scale models at a pace without precedent in the history of general purpose technologies: incremental version numbers arrive every few months, each accompanied by measurable gains on reasoning, coding, multimodal perception, and autonomous task execution benchmarks (Bommasani et al., 2021; Brown et al., 2020). To the ordinary observer, this cadence reads as noise — an endless churn of product announcements in which no single system holds attention for long. This paper takes the opposite view: the churn is the signal. It is precisely what the diffusion phase of a general purpose technology looks like from the inside.

The historical record is instructive. When Paul David (1990) examined why the electric dynamo took nearly four decades to register in productivity statistics, he found that the delay was not technological but organizational: factories had to be physically and procedurally rebuilt around the new power source before its gains could be captured. Something structurally analogous is now occurring with foundation models. The models themselves have crossed critical capability

thresholds; what lags is the reorganization of firms, workflows, professions, and institutions around them. The central claim of this paper is that this reorganization will compress dramatically — that the lag which took electricity forty years will take artificial intelligence fewer than ten, and that the steep portion of the curve begins around 2027.

The paper advances an argument in three movements. First, that the current proliferation of competing models is economically unstable and will resolve into convergence: a small number of dominant, hyper-capable systems operating as shared infrastructure, accessed the way organizations today access search (Google), human capital management (Workday), or cloud computing (AWS) — that is, without ceremony, without wonder, and largely without noticing. Second, that the period beginning in 2027 will exhibit what I term capability-diffusion compression: the interval between a frontier capability existing and that capability being productized, distributed, and absorbed into daily work will shrink toward zero, producing the subjective experience of "something new every day." Third, that the labor-market consequence of this compression is, on net, expansionary rather than contractionary — a position I ground in the task-based framework of Acemoglu and Restrepo (2019) and in the emerging empirical record.

A note on epistemic posture is warranted. Predictive claims about technological trajectories carry irreducible uncertainty, and the history of forecasting is littered with confident errors in both directions (Armstrong & Sotala's caution applies broadly; see also Tetlock, 2005). Accordingly, the claims below are framed as structured hypotheses with stated mechanisms and falsification conditions, not as prophecy. Where the argument depends on extrapolation, the extrapolation is made explicit.

2. Theoretical Framework

2.1 General Purpose Technologies and the Economics of Pervasiveness

Bresnahan and Trajtenberg (1995) define a general purpose technology (GPT) by three properties: pervasiveness across sectors, intrinsic potential for continuous improvement, and the capacity to spawn complementary innovations. Steam power, electricity, and digital computing each satisfied these criteria, and each followed a recognizable arc: an initial period in which the technology was a visible, celebrated novelty; an intermediate period of chaotic competitive experimentation; and a mature period in which the technology became invisible infrastructure — a precondition of economic activity rather than a feature of it. Large-scale AI systems satisfy all three GPT criteria to an unusual degree. Eloundou, Manning, Mishkin, and Rock (2024) estimate that approximately 80% of the U.S. workforce could see at least 10% of their tasks affected by large language models, with roughly 19% of workers seeing half or more of their tasks affected — a pervasiveness profile broader than any prior GPT at a comparable stage of maturity.

The GPT lens yields the paper's first structural prediction: the salience of AI is a phase indicator, not a permanent condition. Technologies are named, discussed, and marveled at only while they are incompletely diffused. Nobody describes their firm as "electricity-powered"; nobody lists "uses the internet" as a corporate differentiator. The very intensity of current AI discourse — the

daily announcements, the version-number theater of successive model generations — marks the technology as mid-transition. The endpoint of the transition is silence: AI as an assumed substrate, a tool in the strict sense, occupying the same cognitive category as the spreadsheet or the search bar.

2.2 Ubiquitous Computing and the Doctrine of Disappearance

Mark Weiser (1991) opened his foundational essay on ubiquitous computing with a sentence that functions as this paper's organizing axiom: “The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.” Weiser was writing about hardware, but the principle generalizes to cognition-augmenting software with even greater force, because cognitive tools are absorbed into the user's own sense of competence. The professional who queries an AI system two hundred times a day does not experience two hundred encounters with artificial intelligence; she experiences her own work, made frictionless. Disappearance, in this sense, is not the failure of a technology to matter — it is the highest form of mattering.

2.3 Recursive Improvement and the Compression of Innovation Cycles

The theoretical possibility that machine intelligence could enter a self-reinforcing improvement loop was articulated by I. J. Good (1965), formalized as the “singularity” by Vinge (1993), popularized by Kurzweil (2005), and treated with analytical rigor by Bostrom (2014). This paper does not require the strong version of that thesis. It requires only the weak, empirically observable version: AI systems are already principal instruments in the design, training, and evaluation of their successors, and AI-assisted software development materially shortens the build cycle of AI-powered products. This creates a second-order compression — not merely faster models, but faster tools built on the models, built partly by the models. Amodei (2024) sketches the plausible near-term consequences of such compounding in domains from biology to economic development. The mechanism matters here because it supplies the engine for the Acceleration Hypothesis developed in Section 4: daily novelty is what a compounding toolchain looks like from the outside.

3. Phase One — Proliferation: The Present Cambrian Moment (2022–2027)

The current period is best understood as a Cambrian explosion in the strict evolutionary sense: maximal experimentation in form, most of which will not survive. Dozens of frontier and near-frontier systems compete simultaneously; enterprises pilot several in parallel; consumers toggle between assistants the way early-twentieth-century households toggled between competing electrical standards. Perez (2002) documents that every technological revolution passes through exactly such an “installation period” — characterized by speculative over-entry, redundant capital formation, and intense public attention — before consolidating into a “deployment period” in which a rationalized version of the technology diffuses through the whole economy.

Three forces make consolidation in AI more probable, and more extreme, than in prior installation periods. First, capital intensity: frontier training runs require compute expenditures accessible to

only a handful of actors, creating natural-oligopoly dynamics at the model layer even as the application layer remains radically open. Second, data-network effects: systems that mediate more of the world's work accumulate feedback that widens their lead. Third, integration gravity: once an organization's workflows, memory, and tooling are built around a given system, switching costs approach those of replacing an ERP platform — which is to say, prohibitive. The equilibrium implied by these forces is the paper's convergence claim: a very small number of super-capable systems functioning as the cognitive utility layer of the global economy, with an immense long tail of specialized tools built on top of them. The structural analogy is exact: many websites, one dominant search layer; many corporate processes, a few dominant systems of record; many applications, a few hyperscale clouds.

4. Phase Two — Acceleration: The 2027 Compression Hypothesis

4.1 Statement of the Hypothesis

The Acceleration Hypothesis holds that beginning approximately in 2027, the elapsed time between (a) a capability existing at the frontier and (b) that capability being embodied in a distributed, usable tool will compress to days or hours. The subjective consequence for practitioners and the public will be a sustained, multi-year period in which genuinely novel instruments — not merely announcements, but working tools that change what an individual can do before lunch — appear at a near-daily cadence. The objective consequence will be the obsolescence of essentially every pre-2024 knowledge workflow: not because the old methods stop functioning, but because their opportunity cost becomes indefensible, in the same way that a 2005-era paper filing system still “works” and yet no serious enterprise would operate one.

4.2 Mechanism

Four gears drive the compression. (1) Autonomous software agents collapse the product-development cycle: what required a funded team and a quarter now requires a specification and an afternoon, so the population of builders expands by orders of magnitude. (2) The marginal cost of intelligence falls along a steep experience curve, as inference costs per unit of capability have historically declined at rates far exceeding classical hardware curves, moving advanced capability from scarce to ambient. (3) Distribution is pre-solved: unlike prior GPTs, which required physical infrastructure to reach users, an AI-era tool reaches billions of devices at the moment of its creation. (4) The recursive loop of Section 2.3 means each of the first three gears is itself being accelerated by the technology it accelerates. No prior technological transition has possessed all four properties simultaneously; this conjunction is the analytical basis for expecting the 2027–2035 window to be qualitatively unlike the electricity or internet transitions in speed, even where it rhymes with them in structure.

4.3 The Discontinuity Claim

It follows that 2024 will function historically the way 1995 functions for the internet or 1913 for electrified manufacturing: as the last legible year of the prior regime. Brynjolfsson and McAfee

(2014) argued that digital technologies had brought the economy to an inflection they called the "second machine age"; the framework of this paper treats foundation models as the completion of that inflection rather than a new one — the moment the second machine age acquired a general-purpose cognitive engine. Concretely, the discontinuity claim predicts that by the early 2030s, (i) unassisted drafting, coding, analysis, scheduling, and search — the core verbs of pre-2024 knowledge work — will persist only as deliberate craft practices, analogous to letterpress printing; (ii) organizational charts will encode human-agent teaming as a default structural unit; and (iii) the phrase "AI-powered" will have vanished from product marketing for the same reason "electric-powered" did.

5. Phase Three — Invisibility: Superintelligence as Utility

The teleology of the argument now becomes explicit. The endpoint of proliferation and acceleration is not a dramatic, singular unveiling of "the super AI." It is subtler and stranger: a system (or small federation of systems) of extraordinary capability that the world addresses the way it addresses Google — hundreds of billions of times a day, with total unselfconsciousness. Agrawal, Gans, and Goldfarb (2018) demonstrated that the economic essence of machine intelligence is a collapse in the price of prediction; when the price of any input collapses far enough, the input stops being a strategy and becomes a utility. Electricity, computation, and connectivity each completed this trajectory. Intelligence — the last and most valuable input — is now visibly on the same path.

This reframing dissolves a false dichotomy that dominates public discourse: the choice between AI as overlord and AI as toy. The historically grounded third option is AI as substrate — a layer that "runs the world" in precisely the sense that electrical grids, TCP/IP, and payment rails run the world: totally, silently, and under human institutional governance. The claim that a super AI will "become just a tool" is therefore not a deflationary claim about its power; it is the strongest possible claim about its power. Tools that reach utility status are the only technologies that ever achieve total adoption. The paper's synthesis can be stated in one sentence: the singularity, if the word is to retain analytical meaning, will not be an event that is watched — it will be an infrastructure that is used.

6. The Labor Question: Why the Task-Creation Channel Dominates

6.1 Two Channels, One Empirical Record

Automation economics distinguishes a displacement effect, in which machines absorb tasks previously performed by labor, from a reinstatement effect, in which technology creates new tasks in which labor holds comparative advantage (Acemoglu & Restrepo, 2019). Public anxiety attends almost exclusively to the first channel — an anxiety given influential quantitative form by Frey and Osborne (2017), whose estimate that 47% of U.S. employment was at high risk of computerization shaped a decade of discourse. Yet the two centuries of evidence surveyed by Autor (2015) show the reinstatement channel systematically dominating over any horizon longer than a business cycle:

the occupational categories employing most workers today did not exist when their grandparents entered the workforce. Bessen (2015) supplies the canonical micro-illustration — the automated teller machine, which reduced the cost of operating a bank branch, expanded the number of branches, and left total teller employment higher for decades while transforming the content of the job from cash-handling to relationship work.

6.2 The Expansionary Case for the AI Transition

Three considerations suggest the AI transition will be more expansionary, not less, than its predecessors. First, demand elasticity: the goods AI makes cheaper — software, analysis, design, education, care coordination, scientific discovery — are goods for which human appetite is effectively unbounded, so cost collapse translates into volume explosion rather than headcount reduction. Second, the democratization of production: when building software, media, or businesses no longer requires scarce specialist skills, the population of economic creators expands from millions to billions, and every new creator is a new employer of complementary human effort. Third, the empirical trajectory already visible: the World Economic Forum's (2025) survey of over a thousand global employers projects 170 million jobs created against 92 million displaced by 2030 — a net gain of 78 million — with the fastest growth precisely in AI-adjacent, data, and technology-diffusion roles. Entire occupational strata now emerging — agent-fleet supervisors, AI-workflow architects, model-behavior auditors, human-AI interaction designers, synthetic-media provenance specialists — have no 2019 antecedents, exactly as the task-creation framework predicts.

6.3 The Honest Caveat

The expansionary case is a claim about net aggregates and end states, not about the smoothness of the path. Acemoglu and Restrepo (2019) are explicit that the balance of channels is a policy variable, not a law of nature; transitions of this magnitude have historically imposed severe localized and generational costs even while enriching the aggregate. The forty years separating the dynamo from the productivity statistics (David, 1990) were not painless years. A responsible statement of the thesis is therefore conditional: the AI transition creates more work than it destroys provided that institutions — education, credentialing, labor mobility, and safety nets — adapt at a speed commensurate with Section 4's compression. That proviso is the true policy frontier of the coming decade.

7. Falsifiable Predictions

A framework of this kind earns credibility only by exposing itself to refutation. The model advanced here fails, and should be discarded, if by 2032: (1) the frontier model layer remains fragmented among ten or more roughly co-equal general systems, contradicting the convergence claim; (2) the median time from frontier capability to mass-distributed tooling has not fallen below thirty days, contradicting the compression hypothesis; (3) aggregate employment in the advanced economies is durably lower than its 2024 level after controlling for demography, contradicting the reinstatement hypothesis; or (4) “AI-powered” persists as a meaningful marketing differentiator,

indicating the technology never achieved infrastructural invisibility. Conversely, the framework is corroborated to the extent that AI discourse itself quiets — the surest sign, on this paper's logic, that the transformation is complete.

8. Counterarguments and Limitations

Four objections deserve direct engagement. The scaling-wall objection holds that capability gains may plateau, stranding the compression mechanism; this is possible, though the historical pattern of GPTs is that even a frozen frontier sustains decades of diffusion-driven transformation — the argument of Sections 3–5 survives a plateau, and only Section 4's daily cadence would soften. The governance objection holds that regulation may deliberately slow deployment; granted, but regulation shapes the path of utility formation rather than preventing it, as the histories of aviation, pharmaceuticals, and telecommunications attest. The concentration objection warns that a convergent cognitive utility poses unprecedented questions of power — this paper concedes the point entirely and regards the governance of the intelligence utility as the constitutional question of the 2030s, while noting that it is a question about a tool, which is itself a confirmation of the thesis. Finally, the forecasting-humility objection: all long-range technological prediction, including this one, inherits the base rates documented by Tetlock (2005). The response is methodological rather than rhetorical — hence Section 7. This paper is additionally limited by its reliance on public data, its focus on knowledge work over physical labor, and its predominantly advanced-economy frame; the diffusion dynamics in developing economies, where the leapfrogging potential is arguably greatest, merit dedicated study.

9. Conclusion

This paper has argued that the correct mental model for the coming decade of artificial intelligence is not the arrival of a rival but the completion of a utility. The present proliferation of models is the noisy installation period of a general purpose technology; the approaching acceleration — dated here to roughly 2027 — is the compression of that technology's diffusion cycle toward zero; and the destination is invisibility: one, or a very few, super-capable systems woven so deeply into economic life that they are experienced not as intelligence but as competence — one's own. Nothing about work will function as it did before 2024, and yet the deepest change will be the least dramatic in appearance: more people employed, in occupations that do not yet have names, using an instrument they no longer think to mention. The most profound technologies, Weiser taught, are those that disappear. The thesis of this paper is that humanity is presently building the most profound one, and that its disappearance has already begun.

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